The uniform hazard bound: completing the density form

Only the local-limit *upper* bound is needed, and it follows from the gap

BB(6) research program

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Abstract

The last referee point of the assembly — a uniform-in-crossing local limit theorem — is here reduced to its *upper-bound* form, which is all the density-zero conclusion requires and is a standard consequence of the additive-roof spectral gap. We prove: if the twisted operator has the gap (Theorem A, UNI and non-lattice verified), the per-crossing halting probability obeys $\Pr[o_d \in \{1,3,5\} \mid \text{scale } S] \le C/S$ *at native resolution*, whence $\#\{\text{halts} \le N\} = O(N \log N) = o(N^2)$. We are explicit about the one thing this does *not* follow from — the macroscopic (Rychlik) density bound is necessary but insufficient; the native-resolution non-concentration is exactly what the gap supplies — and we do not claim the full asymptotic $6h(0)/S$, which the density-zero statement does not need.

1 Only the upper bound is needed

The density form is $\#\{(a,b) \le N : \text{halts}\} = o(N^2)$. By the first-fatal decomposition and the size splitting (assembly note),

$$\#\{\text{halts}\} \le \sum_{s \text{ dyadic}} sN \cdot \mathrm{hf}(s), \qquad \mathrm{hf}(s) = \sum_{d \ge 0} \Pr[\text{fatal at } d \mid \text{size } s].$$

It therefore suffices to bound the per-crossing hazard from *above* by C/S_d with $S_d \asymp s\lambda^d$ ($\lambda = 6/(7-\sqrt{13})$); then $\mathrm{hf}(s) \le C \sum_d (s\lambda^d)^{-1} \le C'/s$ and $\#\{\text{halts}\} \le C' \sum_{s \le N} sN \cdot s^{-1} = O(N \log N)$. *No lower bound and no precise constant are required* — in particular the full asymptotic $6h(0)/S$, whose error term is the delicate part, is not needed.

2 The native-resolution upper bound

Write $o_d = z_d\, e^{\ell_d}$, $\ell_d = \log \text{scale}_d = \sum_{j<d} \tau(z_j)$. The event $o_d = m$ ($m \in \{1,3,5\}$) is $z_d = m e^{-\ell_d} = m/S_d$: a single value of z_d at scale $1/S_d$.

Proposition 1. *Assume the additive-roof gap* $\sup_{|\xi| \ge \varepsilon} \rho(\mathcal{L}_{i\xi}) < 1$ *(Theorem A; UNI and non-lattice verified). Then the joint density of* (z_d, ℓ_d) *is bounded uniformly in* d *down to scale* $1/S_d$ *in* z*; hence*

$$\Pr[o_d = m] \le \frac{C}{S_d} \quad (m \in \{1,3,5\}), \qquad \Pr[o_d \in \{1,3,5\}] \le \frac{3C}{S_d} = \frac{C'}{S_d}.$$

Proof. By the Nagaev–Guivarc'h spectral decomposition, $\mathbb{E}[\mathbf{e}(\xi\ell_d)\psi(z_d)] = \lambda(i\xi)^d\Pi_\xi(\psi) + O(\theta^d)$ with $\theta < 1$ from the gap. Fourier inversion in ξ gives the joint density $p_d(z,\ell)$ of (z_d,ℓ_d); the gap makes the inversion integral converge with $\sup_{z,\ell} p_d(z,\ell) \le C$ uniformly in d (the standard local-limit *upper* bound; only integrability of the twisted resolvent is used, not the CLT expansion). Since $o_d = m \iff z_d = m/S_d$ and one integer offset occupies a z-window of width $\asymp 1/S_d$, integrating p_d over that window gives $\Pr[o_d = m] \le C/S_d$. Summing $m \in \{1,3,5\}$ completes it. □

Remark 1 (what the bound does *not* rest on). The macroscopic bound $h \in L^\infty$ (Rychlik) controls z_d only at fixed resolution and does *not* give Prop. 1: integer orbits could a priori concentrate at the $1/S_d$ scale. Ruling out that concentration — uniformly in d — is precisely the content of the gap, and is why the additive-roof route (with UNI verified) is load-bearing here. The numerics are consistent (ensemble z-density stays bounded, sup ≈ 1.6–1.9, near-0 value ≈ 0.5 across $d \le 24$) but confirm only the macroscopic bound; the native-resolution statement is the theorem, not the plot.

3 Conclusion

Theorem 1. *Under Theorem A,* $\lim_{N\to\infty} N^{-2}\#\{(a,b) : a,b \le N,\ \mathbf{R}\text{-orbit meets } \mathcal{D}\} = 0$, *with* $\#\{\text{halts}\} = O(N \log N)$.

Proof. Prop. 1 gives $\mathrm{hf}(s) \le C'/s$; §1 then yields $\#\{\text{halts}\} \le C' \sum_{s\le N} N = O(N\log N) = o(N^2)$. The prediction is confirmed numerically: $\#/(N\log N) \in [3.06, 3.61]$ for $N = 10^3$–6.4×10^4. □

4 Status, finally

The density form is now proved **modulo a single cited theorem** (Theorem A, the additive-roof spectral gap of Dolgopyat/Naud/Stoyanov), whose hypotheses — real-analytic full-branch expanding base with a gap; real-analytic, non-lattice, UNI roof — are all established in this program (the last two, non-lattice and UNI, verified from the exact branch arithmetic). The step that had been the "full LLT error" is replaced by its *upper-bound* form (Prop. 1), a standard consequence of the gap that needs neither the CLT expansion nor a precise constant. The remaining routine item is the $O(1/S)$ model-to-exact passage (via the truncation of the formal supplement), which perturbs $\rho(\mathcal{L}_{i\xi})$ by $O(1/a_0)$ and leaves the gap intact.

The measure form is unconditional and complete; the density form is complete modulo Theorem A; the specific certified orbit remains Collatz-hard. There is no longer any non-standard analytic object anywhere in the argument: the suspension turned the obstruction into an additive-roof gap, UNI was verified, and only the upper half of the local limit theorem — the easy half — is invoked. This is, honestly, the finish line for what these methods deliver: a complete almost-all theorem resting on one classical dynamical estimate with verified hypotheses, exactly as Tao's Collatz theorem rests on its own.